ORF309/MAT380
Lifetimes I:
Definitions & Series Connections

(pp. 77-84)

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Agenda for today

- Lifetime of components and devices
- Definitions
- Hazard function
- Hazard rate
- Connecting Lifetimes

Definitions

- The term **Lifetime** is perhaps the most important **random variable** of our **lifetime**…

- **Everything has a Lifetime** → L=“the elapsed time until the occurrence of a specific event” (failure, death, etc.)
  - Humans, Cars, Houses, Electronics, Wealth, Power, Friendships, …

- The values of Lifetime can vary
  - L is a **Continuous RV**
  - Has values: $0 < L < \infty$ (the Definition of Lifetime)
  - Has **distribution** (**CDF**): $F(t) = P\{L \leq t\}$, and the **time** is **continuous** $t \in [0, \infty)$
    - **The probability that something dies (or fails) by time t is very useful to know**

Definitions

- It is more convenient to think of the probability:

$$P\{L > t\} =$$

Shifting perspective:
- From probability of failure
- To probability of survival

- The probability that something *survives* (or *lasts*) beyond time t (aka, *Survival function* $S(t) = P\{L > t\}$)

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  - Hence: $P\{L > t\} = 1 - P\{L \leq t\} = 1 - F(t)$
  - Two very important functions
    - $H(t)$: the (Cumulative) **Hazard function**
    - $h(t)$: the **hazard rate**

They both provide a way to mathematically describe the ***deterioration of survival probability as time increases.***

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$$\boxed{P\{L > t\} = e^{-H(t)} = e^{-\int_0^t h(s)ds}}$$

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  The **total accumulated risk** from time 0 up to time t. It's the integral of the instantaneous risk.

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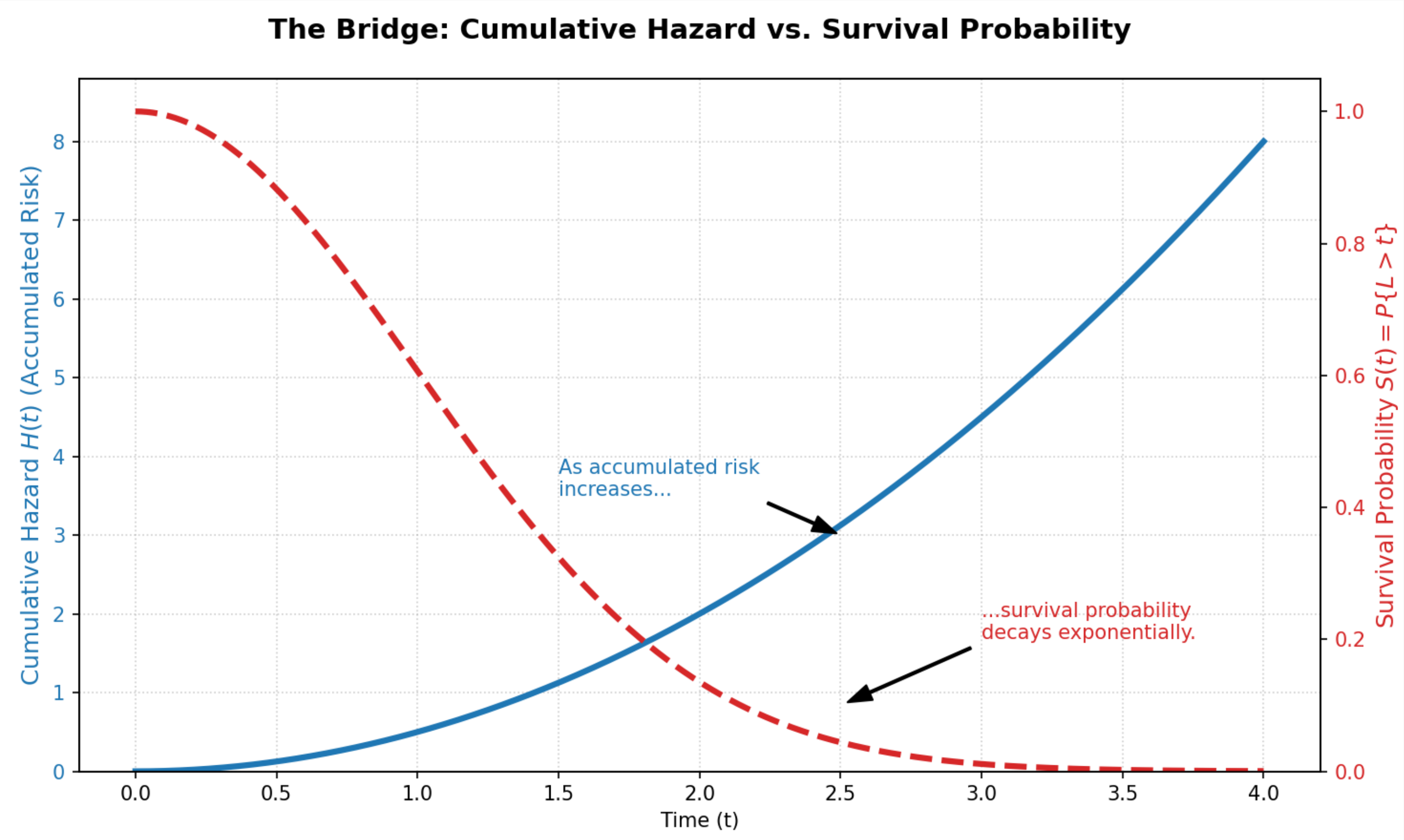
- **Bridge between the two**

$$P\{L > t\} = e^{-H(t)} = e^{-\int_0^t h(s)ds}$$

The probability of survival is **exponentially decaying** as the **total accumulated risk H(t) increases.**



The Bridge: Cumulative Hazard vs. Survival Probability
Cumulative Hazard $H(t)$ (Accumulated Risk)
Survival Probability $S(t) = P\{L > t\}$
Time (t)
As accumulated risk increases...
...survival probability decays exponentially.
0
1
2
3
4
5
6
7
8
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

Properties of Hazard function and hazard rate

- **Properties of Hazard functions** $\boldsymbol{H(t)}$:

1. $P\{L > 0\} = e^{-H(0)} = 1 \Rightarrow H(0) = 0$

   No risk has been accumulated at start

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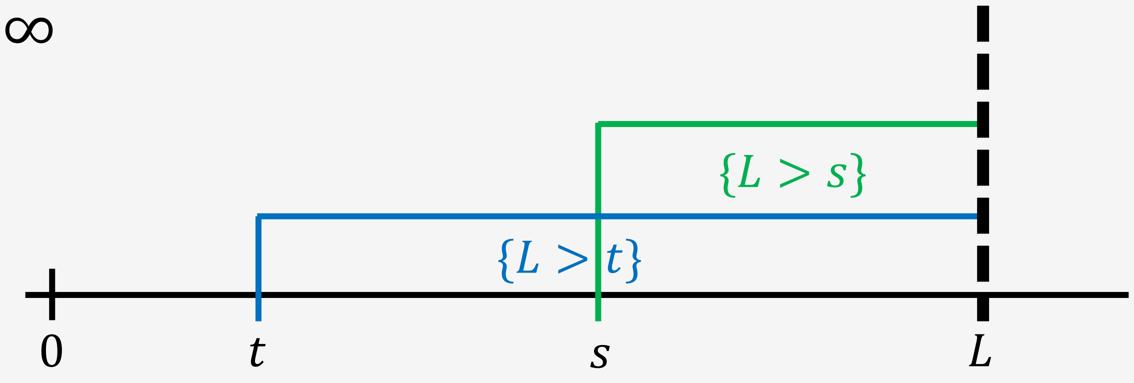
3. $H(t)$ is an **increasing** function

   Risk builds up as we age

   - Take any $t < s$. Then



$$\{L > s\} \subset \{L > t\} \Rightarrow P\{L > s\} < P\{L > t\} \Rightarrow e^{-H(s)} < e^{-H(t)} \Rightarrow H(t) < H(s)$$

Properties of Hazard function and hazard rate

- **Properties of the hazard rate** $\boldsymbol{h(s)}$

  - $H(t) = \int_0^t h(s)ds$ → $h(s) = \frac{d}{ds}H(s)$ (fundamental theorem of Calculus)

  - We know: $H(t)$ is increasing → $h(s) \geq 0$ (nonnegative)

  - Since $H(\infty) = \infty$ → $\int_0^\infty h(s)ds = \infty$ (the integral diverges)

Properties of Hazard function and hazard rate

- **Interpretation**
  - The Hazard function appears in: $P\{L > t\} = e^{-H(t)}$
  - Consider the CRV: $\boldsymbol{X \sim Expon(1)}$. Then the **survival function** of X is:

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$$P\{H^{-1}(X) > t\} = P\{X > H(t)\} = e^{-H(t)} = P\{L > t\}$$

  - So, $\boldsymbol{L}$ **has the same distribution as** $\boldsymbol{H^{-1}(X)}$

Inverse function - *refresher*

Example 1:

Consider a function that converts temperature from Celsius (C) to Fahrenheit (F)

$$F = f(C) = \frac{9}{5}C + 32$$

The inverse function, $C = f^{-1}$, should convert Celsius to Fahrenheit

$$C = f^{-1}(F) = \frac{5}{9}(F - 32)$$

Example 2:

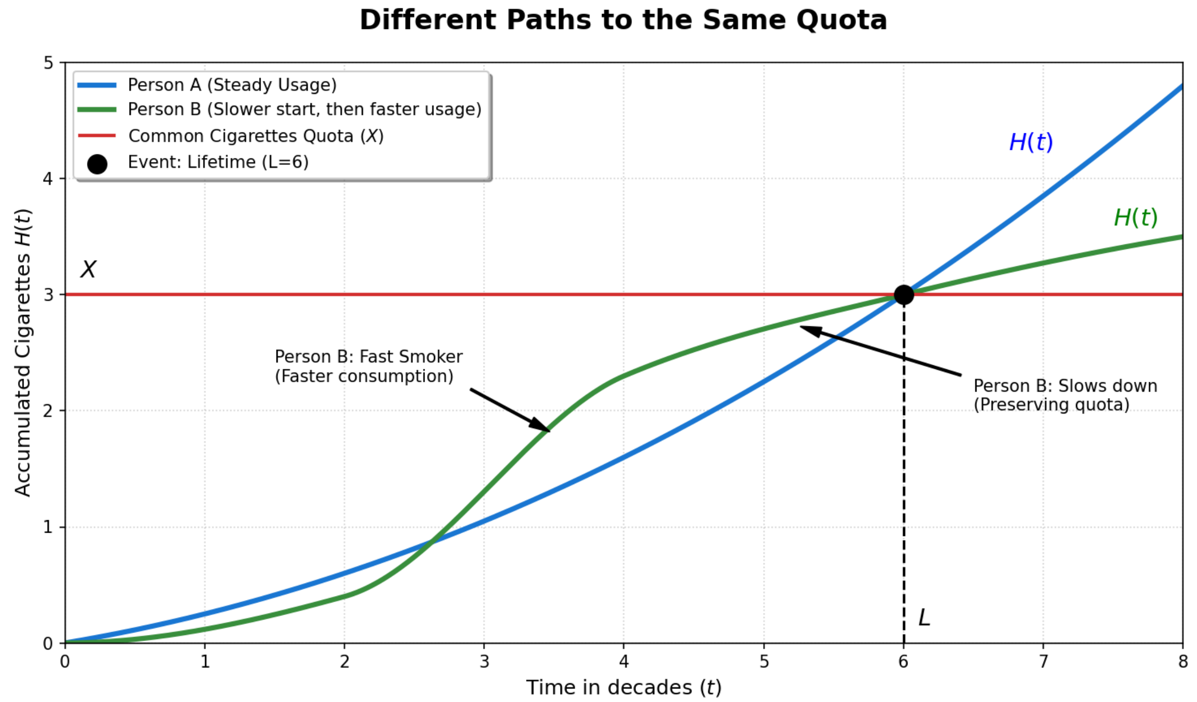
Consider the function: $y = f(x) = e^{-\lambda x}$. We will find its inverse function $x = f^{-1}(y)$

$$\ln(y) = \ln\left(e^{-\lambda x}\right) \Rightarrow \ln(y) = -\lambda x \Rightarrow x = -\frac{1}{\lambda}\ln(y) \Rightarrow f^{-1}(y) = -\frac{1}{\lambda}\ln(y)$$

Properties of Hazard function and hazard rate

Interpretation: Example for better understanding

- $X =$ "total # of **cigarettes** a person can smoke in their lifetime"
  - Different for everyone → $X \sim Expon(1)$
- $H(t) =$ "# of **cigarettes** used up to time t"
  - How fast he consume hs **cigarettes**
- $L =$"time when he use all his **cigarettes**"
  - Once he reaches L , he dies → $L = H^{-1}(X)$
    (he reached his lifetime)



Will you make it another day…?

- **Question**: given you made to t years old, will you leave another u years?

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$$P\{L > t + u | L > t\} = \frac{P\{L > t + u, L > t\}}{P\{L > t\}} = \frac{P\{L > t + u\}}{P\{L > t\}} =$$

$$= \frac{e^{-\int_0^{t+u} h(s)ds}}{e^{-\int_0^{t} h(s)ds}} = e^{-\int_0^{t+u} h(s)ds + \int_0^{t} h(s)ds} = e^{-\int_t^{t+u} h(s)ds}$$

You will not make it another day

- **Question**: given you made to t years old, will you die in the next u years?

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$$P\{L \le t + u | L > t\} =$$

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- **Question**: given you made to t years old, will you die in the next u years?

$$P\{L \leq t + u | L > t\} = 1 - P\{L > t + u | L > t\} = 1 - e^{-\int_t^{t+u} h(s)ds}$$

Now define:

$$F_t(u) = 1 - e^{-\int_t^{t+u} h(s)ds}$$

and note:

$$F_t(0) = 0$$

You will not make it another day

Let's see what happens at the limit:

$$\lim_{u\to 0}\frac{P\{L\le t+u|L>t\}}{u}=\lim_{u\to 0}\frac{1-e^{-\int_t^{t+u}h(s)ds}}{u}=\lim_{u\to 0}\frac{1-e^{-\int_t^{t+u}h(s)ds}-0}{u}=$$

$$\lim_{u\to 0}\frac{F_t(u)-F_t(0)}{u}=-\frac{d}{du}e^{-\int_t^{t+u}h(s)ds}\bigg|_{u=0}=h(t)$$

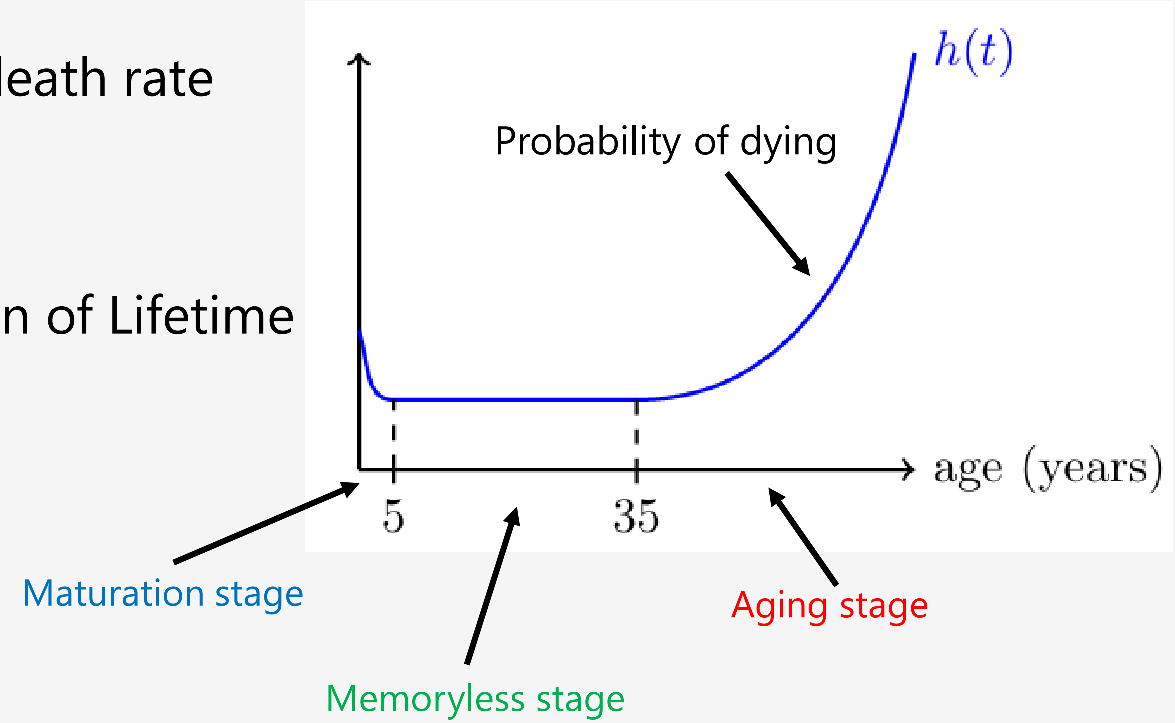
Hence, we have shown:

$$P\{L\le t+\Delta t|L>t\}\approx h(t)\Delta\mathrm{t}$$

Remark: that the hazard rate $h(t)$ can be thought of as the probability of death after Δt years, given that we made it to t years old. → Insurance and Healthcare companies are interested in this!

Graph of Human's hazard rate

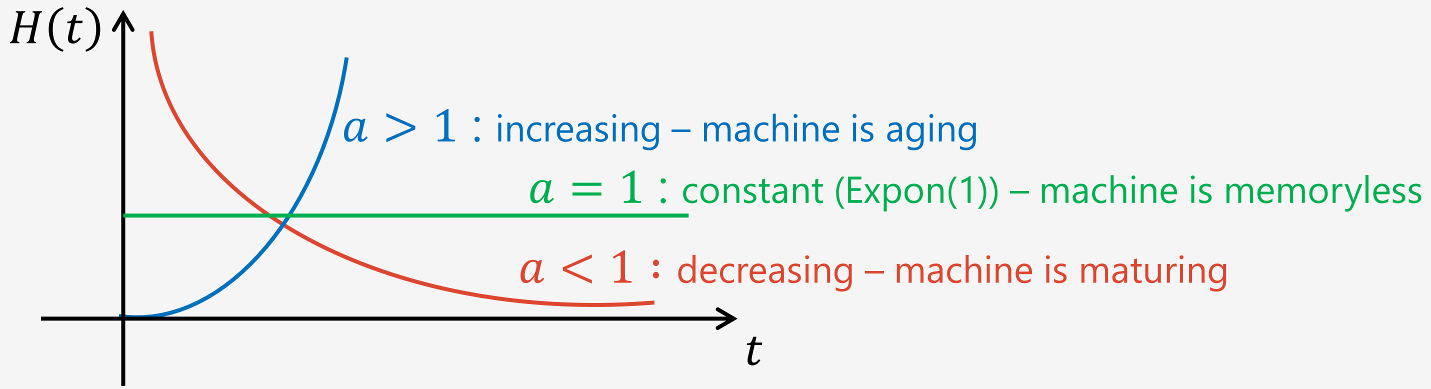
- Infants are vulnerable and have elevated rate of death
- Kids and young humans have the lowest death rate
- As humans grow older than 35-40 years the death rate accelerates
- Hazard rate describes naturally the distribution of Lifetime
  - **Maturation**: when $h(t)$ decreasing
  - **Memoryless**: when $h(t) \approx$ constant
  - **Aging**: when $h(t)$ increasing



Weibull distribution

- **Example 2**:
  - Very useful distribution that can model Aging, Maturation and Memoryless regimes in Lifetimes

$$P\{L > t\} = e^{-\mu t^a} \qquad H(t) = \mu t^a \qquad h(t) = \frac{d}{dt}H(t) = a\mu t^{a-1} \qquad a, \mu > 0$$



Connecting Lifetimes

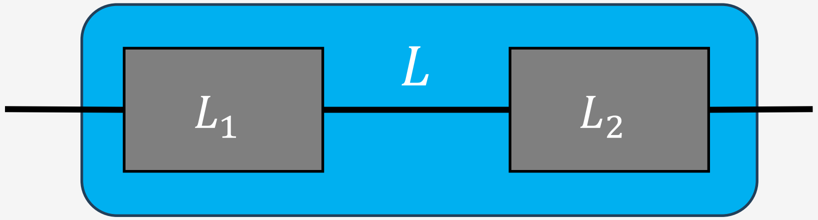
- Connecting Lifetimes is not very applicable to **humans**, (we after may have only one Life; maybe not…) but it is very important for mechanical and electronic **devices**.
  - Many components with different Lifetimes are connected together.
  - What is the Lifetime of the resulting device?
- The **<u>types of connectivity</u>** plays an important role
  - **Serial connectivity**
  - **Parallel connectivity**

Connecting Lifetimes – modeling assumptions

- **Modeling assumptions**
- L_i = Lifetime of **component** $i = 1,2$
  - $L_1 \perp\!\!\!\perp L_2$
  - H_i = Hazard function component $i = 1,2$
- L = Lifetime of the **device**
  - $L = \mathbf{min}\{L_1, L_2\} = L_1 \wedge L_2 \quad \boldsymbol{OR} \quad L = \mathbf{max}\{L_1, L_2\} = L_1 \vee L_2$

Two components connected in series

$$L = \mathbf{min}\{L_1, L_2\} = L_1 \wedge L_2$$



What is the Hazard function H of the device?

$$e^{-H(t)} = P\{L > t\} = P\{L_1 \wedge L_2 > t\} =$$

$$= P\{L_1 > t, L_2 > t\} =$$

$$= P\{L_1 > t\}P\{L_2 > t\} =$$

$$= e^{-H_1(t)}e^{-H_2(t)} =$$

$$= e^{-(H_1(t)+H_2(t))} \quad \Rightarrow$$

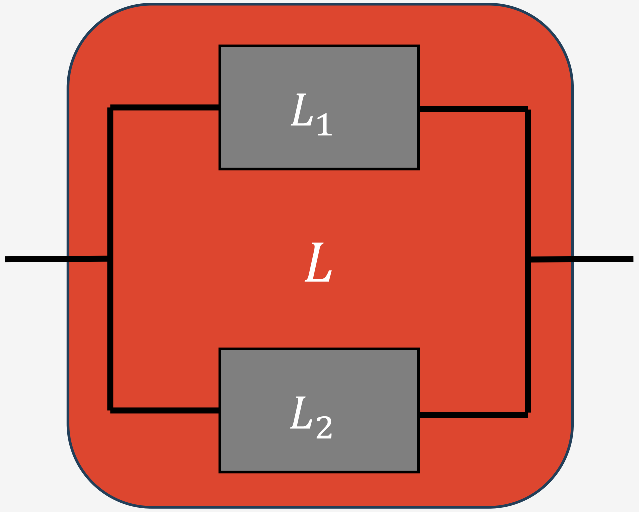
$$H(t) = H_1(t) + H_2(t)$$

Remark: If $X \sim Expon(\mu)$ and $Y \sim Expon(\nu)$

then $\min\{X, Y\} = X \wedge Y \sim Expon(\mu + \nu)$

Two components connected in parallel

$$L = \mathbf{max}\{L_1, L_2\} = L_1 \vee L_2$$



What is the Hazard function H of the device?

$$e^{-H(t)} = P\{L > t\} = 1 - P\{L \leq t\} =$$

$$= 1 - P\{L_1 \vee L_2 \leq t\} =$$

$$= 1 - P\{L_1 \leq t, L_2 \leq t\} =$$

$$= 1 - P\{L_1 \leq t\}P\{L_2 \leq t\} =$$

$$= 1 - \left(1 - e^{-H_1(t)}\right)\left(1 - e^{-H_2(t)}\right) \Rightarrow$$

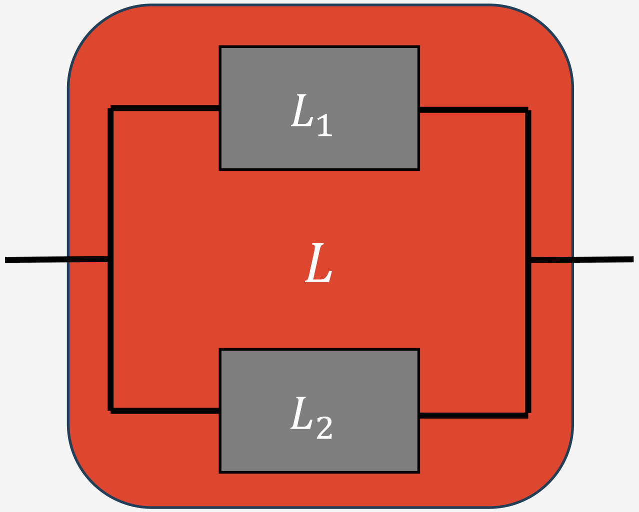
$$H(t) = -\log\left(1 - \left(1 - e^{-H_1(t)}\right)\left(1 - e^{-H_2(t)}\right)\right)$$

Remark: If $X \sim Expon(\mu)$ and $Y \sim Expon(\nu)$

then $X \vee Y$ is **NOT** Exponential

Two components connected in parallel

$$L = \max\{L_1, L_2\} = L_1 \vee L_2$$



What is the **Expected Lifetime** of the device?

Note: $\forall x_1, x_2 \in R,\ x_1 \vee x_2 = x_1 + x_2 - x_1 \wedge x_2 \Leftrightarrow$

$\Leftrightarrow \forall x_1, x_2 \in R, \max\{x_1, x_2\} = x_1 + x_2 - \min\{x_1, x_2\}$

$$E[L] = E[L_1 \vee L_2] =$$

$$= E[L_1 + L_2 - L_1 \wedge L_2] =$$

$$= E[L_1] + E[L_2] - E[L_1 \wedge L_2]$$

Remark: If $L_1 \sim Expon(\mu)$ and $L_2 \sim Expon(\nu)$

$$E[L] = \frac{1}{\mu} + \frac{1}{\nu} - \frac{1}{\mu + \nu}$$

Appendix

Example

Assume a machine has constant hazard rate, say $h(t) = \lambda$,
Note: this machine does not age! → The risk of failure at time $t = 0$ is the same as at $t = 1000$.

Question 1: What is the cumulative Hazard function $H(s)$?

Answer:

$$H(t) = \int_0^t \lambda ds \Rightarrow H(t) = \lambda t$$

Question 2: What is the Survival function: $S(t) = P\{L > t\}$?

Answer:

$$S(t) = P\{L > t\} = e^{-H(t)} = e^{-\lambda t}$$

It represents the survival function of an exponential RV.
It shows how the probability of staying operational decays over time!

Note also: $P\{L \leq t\} = 1 - P\{L > t\} = 1 - e^{-\lambda t}$